



weak-^{*} continuous and differentiable

equivalent to first variation

Differentiability of EOT: General Case

- Differentiability can be studied for general measures too (e.g. Santambrogio (2015), Feydy et al. (2018))
- For probability measures $\alpha, \beta \in \mathcal{P}(X)$, $\text{OT}_\varepsilon(\alpha, \beta)$ is **weak-* continuous and differentiable**
- **Notion of differentiability** (Feydy et al.): $\mathcal{F} : \mathcal{P}(\mathcal{X}) \rightarrow \mathbb{R}$, is said to be differentiable if there exists functional $\nabla \mathcal{F}(\alpha) \in \mathcal{C}(\mathcal{X})$ such that $\forall \xi: \mathcal{F}(\alpha + t\xi) = \mathcal{F}(\alpha) + t\langle \xi, \nabla \mathcal{F}(\alpha) \rangle + o(t)$. [Note: **equivalent to first variation**]
- When viewed as functional of α (or β), the derivatives of $\text{OT}_\varepsilon(\alpha, \beta)$ are given in terms of dual potentials:

$$\nabla_\alpha \text{OT}_\varepsilon(\alpha, \beta) = f \quad \text{and} \quad \nabla_\beta \text{OT}_\varepsilon(\alpha, \beta) = g$$

where (f, g) satisfies $f = T(\beta, g)$ and $g = T(\alpha, f)$ with T the softmin Sinkhorn map:

$$T(\alpha, f)(y) = -\varepsilon \log \int_X \exp\left(\frac{-c(x, y) + f(x)}{\varepsilon}\right) d\alpha(x)$$

- Proof sketch:
 - Optimality conditions for dual pair (f, g) : $f = T(\beta, g)$ α -a.e. and $g = T(\alpha, f)$ β -a.e.
 - Optimal potentials exist, are unique, and satisfy: $\text{OT}_\varepsilon(\alpha, \beta) = \langle \alpha, f^* \rangle + \langle \beta, g^* \rangle$
 - Can be extended from α -a.e., β -a.e. to derivatives defined on entire $\mathcal{X}$ by taking $f = T(\beta, g^*)$ and $g = T(\alpha, f^*)$

Limitations of Exact Derivatives